BEST Hydrodynamics Package
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Introduction

The BEST hydrodynamics package allows for the precise computation of hydrodynamic transport properties of arbitrarily shaped macromolecules subject to the stick boundary condition. In particular, the present version of the package will compute:

1) The coordinates of the Centroid of the Molecule
2) The translational friction **tensor**
3) The translational/rotational coupling friction **tensor** in the Center of Resistance
4) The coordinates of the Center of Resistance
5) The rotational friction **tensor** in the Center of Resistance
6) The Translational diffusion **tensor** in the Center of Mobility
7) The translational/rotational coupling diffusion **tensor** in the Center of Mobilty
8) The coordinates of the Center of Mobility
9) The Rotational diffusion **tensor**
10) A summary of the traces and eigenvalues of the tensors in both internal units (Angstrom, seconds) and cgs units (for water solvent at 20 C). Please see the "Units" section below for an explanation of the units used in Best and how to convert for other units of length, temperatures, or solvents.
11) The Viscosity Factor of the molecule in the centroid or center of viscosity, and if requested, the Instrinsic Viscosity.

The Center of Resistance (CR) is the proper coordinate origin to compute all friction tensors. At this origin, the translation/rotation friction coupling tensor is symmetric. The Center of Mobility is the proper coordinate origin to compute all diffusion tensors. At this origin, the translation/rotation diffusion coupling tensor is symmetric. *The coordinates of the CR are given with respect to the Centroid of the molecule, while those of the CM are given with respect to the CR.* All of these centers coincide for an object with spherical, cubic group symmetry, or axially symmetric objects. They are shifted from one another in the case of a bent rod. See Kim & Karilla, *Microhydrodynamics*, Butterworths, 1991. The typical user will want to focus on the summary section at the end of the output, and convert to his/her solvent and temperature conditions by a simple multiplication as described in the "Units" section below.

The theory and implementation of the computation in BEST are described in detail in the following reference:

S.R. Aragón, "A precise boundary element method for macromolecular transport properties", J. Comp. Chem, **25**, 1191-1205 (2004).

The package does not use a bead representation of the macromolecule, but rather a detailed description of the surface accessible to the solvent is used instead. The solution of the Stokes Creeping Flow equations can be expressed as an exact surface integral over the Oseen Tensor. Consequently, with precise integration over triangular surface elements and regularizing the integral equation, a numerically exact computation of the hydrodynamic transport properties becomes possible. This method is superior to the traditional bead based methods in this regard.

Contents of the Package

The package is provided in a tar file, best.tar. Please untar it in a suitable directory. The package requires the use of 3 programs, and one library.

1) A surface triangulation program
   (Connolly's, MSRoll; http://www.biohedron.com, $500)
   (Sanner's MSMS : http://
   mgl.scripps.edu/people/sanner/html/msms_home.html , free)
2) Rcoal/Coalesce, a triangulation post-processing program (included)
3) BEST, the hydrodynamics program (included).
4) Intel Math Kernel Library (MKL; http://www.intel.com/software/products/);
   or the AMD ACML Library; http://developer.amd.com/acml.aspx
   or the equivalent library provided by your processor's manufacturer.

In addition, the package includes a short reference manual (this document); sample files; Fortran 90 source code and 64 bit binaries to Coalesce and BEST (Linux, Red Hat Enterprise 5.x), and Mathematica packages and notebooks to visualize the triangulated surfaces.

Compiling the source code:

Msroll will easily produce triangulations containing 20000 or more triangles. The larger the macromolecule, the more triangles are produced. The hardware that is used to run the computations must have sufficient memory to hold the arrays of size 3Nx3N for N triangles. The memory requirements double if double precision is used. Larger matrix sizes require double precision because the condition number of the matrix increases rapidly. In addition, the Linux kernel must allow for sufficient stack space to enable the compiler to use large arrays. The file kernel_issues.txt provides some basic information on how to reconfigure a Linux kernel to ensure a large enough stack space.

The source code for coalesce and BEST is written in Fortran 90. Array sizes are allocated dynamically. We use the Intel Fortran Compiler to ensure compatibility with the MKL library. The MKL library is also compatible with Digital Fortran for Windows. (For AMD Opteron, we use the PGI compiler and the AMD ACML library). The following compilation commands give you an idea of the process. I recommend you use the **make** files that come with the package, and modify them according to your detailed installation.

To compile coalesce or rcoal:

fcom –o coalesce coalesce.f index.f –O

To compile BEST:

fcom –o best best.f tgetgr.f triangleint.f matin3.f jacobi.f lapackstuff –O2

"lapackstuff" has to be replaced with appropriate statements to access your LAPACK library. However, the distribution also includes **make** files that will automatically compile BEST using the AMD ACML libraries. You will still need to modify the **make** files to take into account the directory where your libraries are installed.

Web Hydrodynamics Computation: The SFSU HYDRODYNAMICS WICE

We have developed a very user-friendly tool that can be accessed from any web browser to run the hydrodynamics computations described above. We intend to release this tool to the public. A peek at this tool can be accessed at the url: esmeralda.sfsu.edu/~wice. Please inquire from the author if you are interested in using this tool. The development of this tool is a collaborative project with Prof. Ilmi Yoon, (SFSU Computer Science) and her students.

How to Perform a Computation: On the Command Line

The starting point of a computation is a molecular structure in PDB file format. Such a structure can be obtained from a crystallographic database, or from the results of a molecular modeling project. The sample file provided is the crystal structure of a small protein, lysozyme.

Note that files produced from molecular modeling studies or otherwise having hydrogen atoms included should be pre-procesed to remove the hydrogen atoms. Reason: the hydrodynamic hydration value was obtained without hydrogen atoms. A small Linux utility called hdel is included to remove the H atoms from a pdb file.

Step 1: Generate and Triangulate the Molecular Surface (or Hydrated Surface)

The BEST suite supports two programs for the generation of triangulated surfaces, Connolly's **msroll**, and Sanner's **msms**. A basic comparison of these programs follows.

Differences between msms and msroll:

Cost: msms is free, msroll costs $500

Analytical surface: msroll produces both analytical area and volume for the smooth surface that represents the hydrated surface of the macromolecule. Connolly achieves this by fitting polynomials to the points generated by the rolling sphere. Sanner gives us an analytical surface, but not a volume. Thus, in msms we have only the approximate polygonal value of the volume. This distinction is only important for intrinsic viscosity calculations where an accurate volume is explicitly used. For diffusion tensors, only the analytical and polygonal areas are used, so msms is OK.

Triangulation quality: msms produces triangles that are closer to equilateral, thus producing better triangulations over all. There are no significant differences with properly extrapolated values of the diffusion properties, however.

Hydration control: msroll allows the construction of fairly arbitrary rules for what atoms are inflated and how much. This is essential in the implementation of differential hydration models. msms does not have an easy way of making such distinctions and is best used for uniform hydration models.

USING MSROLL: Connolly's MSRoll program accepts command line arguments in order to perform a computation. The basic flags are:

-m filename.pdb ; the name of the pdb structure file.
-p 1.5 ; the probe radius, typically 1.5 A for water.
-r radius_file ; the name of the file containing the atomic radii.
-t filename.c3p ; the name of the output file containing the triangulation (Coalesce assumes the terminator is .c3p)
-v filename.c3v ; the name of the Connolly output file containing the precise area information. This is required for automatic area correction (regularization).
-f fineness ; the parameter f roughly controls the number of triangles. Larger values produce fewer triangles. A minimum value appears to be around 0.3 and the maximum is 1.5. Try around 1.0 first.

A complete command line for the lysozyme sample file 6YLZ.pdb would be:

msroll –m 6YLZ.pdb –p 1.5 –r radius1.1 –t 6YLZ.c3p –f 0.7 –v 6YLZ.c3v

msroll also produces a record on the screen, which contains a summary of the geometric information, including the precise molecular area and volumes. You can pipe this to a file, but please don't use the .log termination because BEST uses it already. Note that the –r flag is typically used for a hydrated molecule, or when hydrogen atoms are present in the pdb file. See the discussion below.

USING MSMS: Please see the document:
How to run Sanner MSMS.pdf for a description of the multistep procedure to convert a pdb file to a coordinates + radius file, hydrating the molecule, and making a triangulation similar to that produced by msroll.

Biomolecule hydration issues: The radii file must take into account hydration water. Proteins and nucleic acids require different treatment. For proteins we have found that a uniform layer of hydration around the entire molecule describes the hydrodynamic transport properties very well. The sample radius1.1 file contains a list of atomic radii that have been inflated by 1.1 A (this quantity has been added to all radii) in order to account for a little less than half water molecule thick hydration layer uniformly distributed around the protein. This value has been tested for set of non-membrane proteins from small to large molecular weight. The attached Table demonstrates the variance obtained for the translational diffusion coefficient. We believe that most of the variance comes from the experimental values because BEST is very accurate (and some of the values we found for proteins are very old). This work is described in detail in the following publications:

S.R. Aragon and D. K. Hahn "Precise boundary element computation of protein transport properties: Diffusion Tensors, specific volume, and hydration", Biophysical Journal, **91**:1591-1603 (2006).

D. K. Hahn and S.R. Aragon, "The intrinsic viscosity of proteins from high precision boundary element calculations", J. Chem. Theory & Comput. ***2***: 12-17 (2006).

For DNA, the water of hydration is not uniformly distributed around the surface, but appears to be concentrated in the minor groove, and to a lesser extent, in the major groove. A way to simulate this is to inflate the nitrogen atoms by about 2.0 A, and the rest of the atoms by about .2 A. This work is described in the following MS Thesis:

T. Takeda, "DETERMINATION OF WATER HYDRATION LOCATION IN DNA OLIGOMERS THROUGH DIFFUSION COEFFICIENT CALCULATION ", M.S. Thesis, SFSU, August 2005.

After the completion of Step 1, you will have a .c3v and a .c3p file containing geometric information and a surface triangulation. For maximum accuracy, the program performs the "area correction" and it will need the precise value of the hydrated surface area (or molecular surface area if no solvation occurs). The area correction involves a rescaling of the integrals over the Oseen Tensor to compensate for the fact that a triangulation of the molecular hydrated surface has an area smaller than the precise smooth surface. However, the area correction is actually also a regularization process of the ill-conditioned integral being solved. It should not be omitted for amorphous objects.

Step 2: Post process the triangulation with Coalesce

The triangulations produced by the majority of available programs are not suitable for boundary element calculations as is. The triangulations contain copious amounts of small area and very slender triangles that must be eliminated without leaving any "holes" on the triangulated surface. In addition, the programs produce very large number of triangles with unnecessary detail that prolongs the hydrodynamic computation. The subroutine Coalesce fixes the small and slender triangle problems and reduces the triangle count. Coalesce is called via the regular program **rcoal**. There are two versions of rcoal, Version 2.0 reads triangulations produced by Connolly's msroll, while rcoal3 (Version 3.0) can read triangulations from both msroll and Sanner's msms program. A usage statement will be produced if one executes the program without flags:

USAGE, Version 2.0: typing "rcoal" without arguments produces the usage statement:

```
'Usage : rcoal -f connolly_filename.c3p '
'       [-nmax Nmax] [–nmin Nmin] [–n m]
```

USAGE: Version 3.0: typing "rcoal3" without arguments produces the usage statement:

```
SFSU, Aragon Lab.
 Usage: rcoal -f filename -nmax val -nmin val
 -n numb [-s]
 Usage: rcoal -f filename -e val [-s]
```

In rcoal3, the flag –s is used to process msms triangulations.

Coalesce can be used to prepare m files with a number of triangles that varies between Nmax and Nmin for a computation that will extrapolate to an Infinite number of triangles.

If values for m, Nmax, Nmin are missing, then Coalesce does a single calculation to produce an unpredictable number of triangles, but typically a few thousand. For each value of N between Nmax and Nmin, Coalesce produces a log file (filenameN.his) with a histogram of the edge lengths in the modified triangulation, along with statistics about the numbers of triangles eliminated. In addition, coalesce produces an output file containing the triangle vertices in a format required by BEST. The name of the output file is the name of the input file with any endings striped off and the value of N appended. Furthermore, the molecule area and volume information is read from the Connolly .c3v file and written on the first line of the input file. The initial set of comments in the Fortran 90 source code provide extra information.

A useful heuristic to decide the value of nmax is the formula: nmax = 30 $M^{.5}$ = 30.Sqrt[M], where M is the molecular weight in g/mol. The extrapolation graph should look linear for the most accurate results. If the graph is not linear, then discard the smaller number of triangles, add a new larger value to the set, and repeat the extrapolation. The care in this step is highly dependent on how much accuracy you are interested in. You can reduce the nmax value if your extrapolation graph is close to linear, but avoid trying to fit to a quadratic – that actually increases the uncertainty in the intercept.

To process the lysozyme example, one would use the following command:

rcoal –f 6YLZ.c3p – Nmax 7000 –nmin 2500 –n 4

The output files are **6YLZ2000.his** and **6YLZ2000**, **6YLZ3000.his** and **6YLZ3000, 6YLZ4000.his** and **6YLZ4000, 6YLZ5000.his** and **6YLZ5000**. Best uses the **6YLZ[N]** files as input, where [N] is the number of triangles in the file. The actual number of triangles will be close to the numbers given in the example, but not identical to them. A conversion algorithm between the requested number of triangles and the internal control parameter of coalesce is used to approximate closely to the number of requested triangles. Coalesce actually does two extra calculations to produce fit parameters for this internal conversion process. A typical run of coalesce for the above in a modern machine takes only a few minutes.

The program uses an algorithm that scales as the square of the number of vertices. Computation progress messages are displayed on the screen. The present version of coalesce is fairly robust, however, if coalescence should fail, one can usually work around it by re-running the msroll with a different fineness parameter and then trying coalesce again.

A typical histogram file is shown in the appendix.

Step 3: Visualize the Triangulation with Mathematica [Optional]

Especially in the beginning, it is very instructive to visualize the triangulations produced by msroll or coalesce. A very convenient tool for doing this is Mathematica by Wolfram Research. If you do not have this program, please visit http://www.wolfram.com for further information. The included Mathematica notebook demonstrates how to visualize the triangulations. Place this file in the directory where you have the triangulation files and change the SetDirectory command in the notebook to refer to this working directory. Open the notebook with Mathematica (version 3 or above). Note: Version 4.x or later is

assumed. The initial command to rotate the 3D graphics with the mouse will not work in previous versions.

Step 4: Run BEST

BEST is a very precise implementation of the Youngren-Acrivos hydrodynamics on the surface formulation. The input file produced by coalesce contains the coordinates of the vertices of a surface triangulation for the molecule of interest. The present version performs computations with the "stick" boundary condition. The solution of the dense large linear system of equations scales as the cube of the number of triangles and is very computationally intensive. With modern 1 GHz Pentium or faster processors, the computations on a protein take 10-20 min (we use multi-core AMD Opteron or larger processors with for optimum job throughput. There is no significant penalty for double precision on these systems.)
The typical command line for BEST is:

best –f 6YLZ2000 -v &

Typing "best" without arguments prints out a usage statement:
Usage: best –f input_filename [-na] [-v] [-vc] [-mw wm] [-p]

Best Flags and filename appends:
-na Omit the area correction (this is actually a regularization, so it is not advised!)
Append "n"
-v Compute the Viscosity Factor (in the Centroid)
Append "v"
-vc Compute the Viscosity Factor in the Center of Viscosity (longer calculation).
Append "vc"
-mw Compute the Intrinsic Viscosity. Type the value of the molecular weight, wm, in g/mol, omitting the units. Append "m"
-p Perform a pre-averaged hydrodynamic interaction calculation.
Append "p".

Best output naming scheme:
Best will append a sequence of letters to the input filename indicating which flags where set for the calculation, and add a .be ending. The appended letters are given above after the word "Append".

Best performs the area correction by default, using the accurate hydrated surface area obtained by msroll from the .c3v file. Computation progress is written to a log file: **6YLZ2000v.log** in this example. Timing information is given in this file. The output file in this case would be named **6YLZ2000v.be** and is provided in the distribution as an example. The –p option implements a preaveraged hydrodynamic calculation –it does not save time- used only for hydrodynamic research purposes.

Potential problems:

At compilation time: Please see the file Huge_arrays...doc for hints on how to ensure that your Linux kernel can handle large arrays. If the program refuses to compile, it is probably due to array size limitations, either due to insufficient physical memory or

insufficient stack space. If you can compile for smaller array sizes, then you have demonstrated that this is the problem.

At run time: It is possible that you can compile a program but the program will give cryptic messages and quit immediately when executed. In this case, you probably have exceeded the stack space and need to reduce your array sizes. Use coalesce with a larger edgfrac to reduce the number of triangles to a manageable size. If you are running the single precision version in order to save memory, then you may find an output file that contains "NaN" or other indication of machine overflow. This will happen if you are trying to do a computation with around 3000 triangles or more. In this case, the only option is to use the double precision version and load your machine with more memory!

In the Results: Inversion of large dense matrices incurs penalties due to round off error. In single precision, a limit of around 1000 triangles is reasonable for a 32 bit processor (Pentium III, Pentium IV). If you are generating a single precision version of the program (double precision is provided), please make sure you change the source code to refer to the single precision LAPACK routines. *Single precision is not at all recommended!* The double precision version will suffer from round off error when more than 5000 triangles are exceeded in a 32 bit processor. To perform computations with larger number of triangles, a 64 bit processor (Intel EM64, Opteron) is recommended in double precision mode. However, this does not indicate that 32 bit machines are too limiting. We have found that the graph of the tensors vs 1/N is very linear (a benefit of the area correction!) over a large range of triangle numbers and can be accurately extrapolated from 3-4 calculations. This will yield a precision better than 1%. A single unextrapolated computation will often be accurate to 5% or better, an accuracy comparable to the experimental error in the measurement of protein diffusion coefficients. In addition, using the average slope of extrapolated lines from similar molecules yields precision of around 1-2 % from a single computed point.

Units

The units chosen internally recognize that a hydrodynamic computation for a *rigid* body is universal for any temperature or viscosity once the size and shape of a *rigid* molecule are given. Thus, a computation does not have to be repeated, because of a change in temperature or viscosity, *if the size and shape are not expected to change*.
If this is not the case, then a new structure must be obtained at the new temperature and solvent and the calculation must be repeated. For small changes in temperature or viscosity, this should not be the case for molecules such as proteins. Then, the actual experimental conditions can be established at the end. This is a convenient assumption but it does not impose limitations of any kind. Give BEST the correct structure for a given experimental situation and the output is accurate diffusion coefficients.

Typical pdb files have the atomic coordinates expressed in Angstrom (A). BEST makes this assumption. Furthermore, BEST factors out the quantity $1/8\pi\eta$ from the Oseen Tensor while it is performing the hydrodynamic computations. These assumptions lead to a set of internal units in BEST where a flow velocity is in A/s and an angular rotation speed is in rad/s. The internal units of all the quantities are thus:

Friction (Resistance)	Tensor	Best Units	CGS	Conversion factor	Value (20C, water)
Translational friction	Ktt	A	g/s	$8\pi\eta$ /(10^8 A/cm)	$2.5181 10^{-9}$ g/(A s)

Trans/Rot friction	Krt	A^2	g cm/s	$8\pi\eta/(10^8 A/cm)^2$	$2.518\ 10^{-17}$ g cm/(A^2 s)
Rotational friction	Krr	A^3	g cm^2/s	$8\pi\eta/(10^8\ A/cm)^3$	$2.518\ 10^{-25}$ g cm^2/(A^2 s)
Diffusion (Mobility)					
Translational diffusion	Dtt	1/A	cm^2/s	$(kT/8\pi\eta)\ (10^8 A/cm)$	$1.607\ 10^{-5}$ A cm^2/s
Trans/Rot diffusion	Dtr	$1/A^2$	cm/s	$(kT/8\pi\eta)\ (10^8 A/cm)^2$	$1.607\ 10^{3}$ A^2 cm/s
Rotational diffusion	Drr	$1/A^3$	1/s	$(kT/8\pi\eta)\ (10^8 A/cm)^3$	$1.607\ 10^{11}$ A^3/s
Viscosity Factor	ξ		dimensionless		
Intrinsic Viscosity	$\xi\ V_p\ N_o/Mw$		cm^3/g		

Three quantities are required to transform these internal units to typical laboratory units. The conversion factors are given here for "cgs" units. In practice, one can use simple ratios to convert the BEST output to a new temperature or viscosity. For example, Dtt(T) = Dtt(To) (T/To) (η(To)/η(T)). For the Viscosity factor, we use the definition, $\xi = [\eta/\eta_0 - 1]/(V_p N)$, where *N* is number/$cm^3$, and V_p is the molecule volume. To compare with experimental values in cm^3/g, use the relation, $[\eta] = V_p N_0 \xi/Mw$, where N_0 is Avogadro's number, and Mw is the molecular weight. When the –mw flag is used, BEST reads the molecular weight from the command line and the actual Intrinsic Viscosity is computed.

Different length scale used in the input file:

BEST uses the units supplied on input as the measure of length. Since a typical input originates from a crystallographic database and these use Angstrom units, the default labeling in the best output uses the symbol A. If a length scale other than A units is used in the input file, all one has to do is to substitute the symbol for the system of units used for A in the expressions in the above table. For example, if one uses mm, then Dtt is in 1/mm in internal units, and the conversion factor to cgs laboratory units is $(kT/8\pi\eta)\ (10$ mm/cm).

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APPENDIX

Memory requirements for a given number of triangles

The main memory usage of BEST is in the storage of the 3Nx3N matrix of Oseen Tensor integrals. Since the computations are done in double precision, each real number uses 8 bytes. Therefore, the memory estimate in GB, for N in thousands of triangles is:

Mem = 72 (N^2/1000) (1000/1024)^3 GB = 0.067055 N^2 GB

In practice, we subtract 0.5 GB for the operating system use to obtain values in the following table:

Max Triangles

GB Mem	*BEST*
4	7225
6	9057
8	10576
12	13096
16	15204
24	18721

Typical histogram output file

```
COALESCE PROCESS RESULTS -Version=    2.500000

 Input File (Connolly): myo.c3p
  Edge fraction =   0.6000000
 Initial number of triangle records= 4860
 Number of degenerate triangles, pass 1= 0
 Number of small triangles deleted, pass 1= 2598
 Number of polygons generated, pass 1= 2262
 total area, pass 1 =    5771.638

 Number of small triangles deleted, pass 2= 1
 Number of polygons generated, pass 2= 1430
 total area, pass 2 =    5718.716
 Average Triangle Area (A^2)= 3.999102
 Average edgelength (A)= 2.585172
 Cut-off length (A) = 1.551103
 Found: 9 slender obtuse triangles, processed: 9

 Histogram bins and count follows:
 Point distribution around  0.3999102 A^2
  0.0000000E+00          0
  7.9982050E-02          0
  0.1599641              0
  0.2399461              0
  0.3199282              0
  0.3999102              0
  0.4798923              0
  0.5598744              0
  0.6398564              0
  0.7198384              0
  0.7998205              0
 Point distribution around   3.999102    A^2
  0.7998204              0
   1.119749              7
```

1.439677	35
1.759605	91
2.079533	119
2.399462	117
2.719390	132
3.039318	104
3.359246	96
3.679174	191
3.999102	73
4.319031	53
4.638959	45
4.958887	60
5.278815	37
5.598743	32
5.918672	36
6.238600	24
6.558528	21
6.878456	0
7.198384	138